Python

Day-wise Notes

Contents

Day 43 · virtual environment in python 2

Day 44 · import 3

another way to import only specifice library . . . 3

you can name the library when required to make it the way you want . . . 3

dir function . . . 3

we can import function or variable from the same program as well . . . 3

Day 45 · how if __name__ = "__main__" works in python 4

Day 46 · os module 5

Day 48 6

Day 49 · file I/O 7

open a file . . . 7

modes : . . . 7

Day 50 · readlines method 8

Day 51 · seek() 9

Day 52 · lamda function 10

Day 53 · Map Filter Reduce 11

Day 54 · is vs == 12

Day 55 · rock paper scissor 13

DAY 43

virtual environment in python

enviornmetn which is used when working on different projects should become easy

this gives you a access to install the version that is different from globaal version

to do this make a folder

1. open the folder in terminal :python3 -m venv (name of Environment)

   this takes you inside the folder

2. (name of Environment)/bin/activate this simply activates the environment

3. deactivate ==> this one to deactivate

pip freeze ---> this tells how many of version are there in python

pip freeze > requirements.txt ---> this writes all teh packages in txt file found outside the folder

pip -r requirements.txt ---> this installs all the version in your computer

for windows :=========>

CODE

```
"""
----------------------------------------------------------------------
python -m venv venv

venv\scripts\activate

to deactivate simply deactivate

----------------------------------------------------------------------"""
#
```

DAY 44

import

import takes all the built in function in python

import (name of library)

CODE
```
import math
```

another way to import only specifice library

from math import pi , sqrt

this installs only pi and sqrt function from math

CODE
```
from math import pi , sqrt
```

you can name the library when required to make it the way you want

import numpy as np ----> numpy is named as np

now everywhere you can use it as np not numpy

CODE
```
import math as m
```

now use this name

CODE
```
result = m.sqrt(9) #---> this gives 3 as output
```

dir function

this function tell or lists all the function in the library

CODE
```
print(dir(math))
```

we can import function or variable from the same program as well

just see go to video sample

CODE
```
# from sample import *
from sample import raja

# print(sample)
print(raja)# this is a function
print(raja())# returns data type as well as name as print statement
```

DAY 45 how if __name__ = "__main__" works in python

CODE

```
from sample import rakesh

print(rakesh())
# rakesh() now if this is not in if __name__ = "__main__"=====> this prints rakesh ed techaer 2 times if inside this file
# prints it one time only if imported in the other file if in the same file it makes the function run

print(__name__ )# tells you from where is this file imported or if inside the same file it shows __main__
```

DAY 46

os module

os module: helps you to create data folders, files , etc

rename many files etc explore this on internet

and after you print the statement then you the folder is created

CODE

```
import os

#please don't run this code else folder named : 48-100 will be created
for i in range(47,100):
    print(os.mkdir(f"data/{i}.py"))
```

os module:

renaming a file:

CODE

```
import os

os.rename("Oldfile.txt","new_file.txt")
```

DAY
48

CODE

```
a = 5

def hello():
    a=6#local variable
    print(a)


hello()
print(a)

def new():
    #modify global variable
    global a #global variable changed
    a=10
    return a


print(new())#function in which global variable value is changed
```

DAY 49

file I/O

file I/O :handling a file

open a file (1. txt method , 2.Binary Method) , read , write , append ,close the file ==> some manipulative commands

open a file

CODE
```
a = open("sample2.txt", "r")  # if not written 'r' by default it is the same
```

modes :

read mode : this gives you content in terminal , if file do not exists throws error

CODE
```
read = a.read()
print(read)
a.close()
```

write mode : this file allows you to write inside the file , if not exists this creates one file

<==========this mode just removes the earlier content and only keeps your new content==========>

CODE
```
c = open("sample.txt", "w")
c.write("Hello World!")
c.close()
```

append(): this adds the content when ever required

CODE
```
d = open("sample.txt", "a")
d.write("Hello World!")
```

create : this creates the file if already exists then throws error

t and b ==> text and binary i.e. open file as txt i.e by default and binary form i.e. pdfs, images etc

CODE
```
b = open("Locationoffile.txt", "rb")  # here r ---> read and b---> binary mode
```

shortcut method:

CODE
```
with open('sample.txt', 'w') as a:#any method can be used in this
    a.write("hello World!2")
```

DAY 50

readlines method

readlines method: this simply read the code line by line

we have to use loop for this

CODE

```
a=open('sample2.txt', 'r')

while True:
    if not a.readline():
        break
    print(a.readline())
```

another using variable

CODE

```
c=open.readlin()

while True:
    b=a.readline()
    if not b:
        break
    print(b)
```

writeline()

CODE

```
f= open('sample2.txt', 'w')
lines=['hello\n', 'World\n', '!\n', "variable\n"]
f.writelines(lines)
f.close()
```

DAY 51

seek()

seek(): this function directly jumps to the given byte leaving then reads the other bytes

CODE

```
a = open("sample2.txt", "r")

p = a.seek(5)  # jumps 5 charecters forward

print(a.read(4))  # this will read 4 charecters forward of the given text
```

tell(): this function tells from where is reading going to start

CODE

```
k = a.tell()  # this will return 5 as output
```

truncate(): with the help of this you can directly fix the size of the file even more charecters are there written or appended

CODE

```
l = open("klo.txt", "w")
o = l.write("Hello World!")
l.truncate(5) #this directly only pics 5 charecters they are hello
```

DAY 52

lamda function

lamda function : this is a way to create mini function in you program

CODE
```
def sum(a, b):
    return a + b
```

this function can be also written as

CODE
```
double = lambda x: x * 2
# name = lambda arguments : operations/Expressions

cube = lambda y: y**3
print(cube(3))
```

this is mostly used when lambda function is passed as an arguments example

CODE
```
def final_output(fx, v):  # here fx keyword is for function
    return v + fx(v)


print(final_output(cube, 3))
```

DAY 53

Map Filter Reduce

map :

CODE
```
cube = lambda x:x**3

lst=[1,2,3,4,5,6]

c=list(map(cube ,lst))#
print(c)
```

filter : this simply filter the element according to the required conditions

CODE
```
def required_value(k):
    return k>2

d=list((filter(required_value , lst)))
print(d)
```

reduce :

1. we need to import reduce this is not built in

this may be used in libraries like numpy pandas

CODE
```
from functools import reduce

list=[1,2,3]

# syntax : variable_name = reduce(lambda arguments : operations)

c = reduce(lambda x,y: (x*y) , list)+10
print(c)
```

mostly we use this in numpy as an to maximize output in an array

DAY 54

is vs ==

is compares the location in memory

== compares the value

is returns true whenever there is a constant just like number , tuple

DAY 55 rock paper scissor

CODE

```
import random

try:
    while True:
        user_inp = input(
            "Enter \n1 for Rock \t\n2 for scissor\t\n3 for paper\t\n4 for Exiting\t\nEnter Your
Choice>>>>>>>>>\n"
        )
        if user_inp.isdigit():
            b = int(user_inp)
            if b == 1 or b == 2 or b == 3:
                r = random.randint(1, 3)
                # for rock as input
                if r == 1 and b == 1:
                    print("Draw\n\n")
                elif r == 1 and b == 2:
                    print("Win\n\n")
                elif r == 1 and b == 3:
                    print("Lose\n\n")
                # for scissor as input
                if r == 2 and b == 1:
                    print("Lose\n\n")
                elif r == 2 and b == 2:
                    print("Win\n\n")
                elif r == 2 and b == 3:
                    print("Draw\n\n")
                # for paper as input
                if r == 3 and b == 1:
                    print("Win\n\n")
                elif r == 3 and b == 2:
                    print("Lose\n\n")
                elif r == 3 and b == 3:
                    print("Draw\n\n")
            elif b == 4:
                print("<<<<<<<<<<<<See You Next Time>>>>>>>>>>>>")
                break
            elif b > 4:
                print("Enter Integer as\t\n\n 1 , 2 , 3\t")

except:
    print("<----Restart the Game--->\t")
```

TABLE

input		1	2	3
rock	1	D	W	L
scissor	2	L	D	W
paper	3	W	L	D

the cleaner way for this is to make logic instead of the using numbers just like b==r i.e. draw then the three case in which i win keep them in or and use else to get other outputs

give your output to ai and then see what are the mistake if it could be the professional generate the code by youself first